

torch.nn.functional.affine_grid#

[https://pytorch.org/docs/stable/generated/torch.nn.functional.affine_grid.ht](https://pytorch.org/docs/stable/generated/torch.nn.functional.affine_grid.html)

Description:

Generate 2D or 3D flow field (sampling grid), given a batch of affine matrices theta. This function is often used in conjunction with `grid_sample()` to build Spatial Transformer Networks . theta (Tensor) – input batch of affine matrices with shape $(N \times 2 \times 3N \times 2 \times 3N \times 2 \times 3)$ for 2D or $(N \times 3 \times 4N \times 3 \times 4N \times 3 \times 4)$ for 3D size (torch.Size) – the target output image size. $(N \times C \times H \times W \times N \times C \times H \times W)$ for 2D or $N \times C \times D \times H \times W \times N \times C \times D \times H \times W$ for 3D) Example: `torch.Size((32, 3, 24, 24))`

Function Signature

```
torch.nn.functional.affine_grid(theta, size,
align_corners=None)[source]#
```

Parameters

Returns

Return type

Returns:

output (Tensor)

Notes & Warnings

NOTE

Note This function is often used in conjunction with `grid_sample()` to build Spatial Transformer Networks .

⚠ WARNING

Warning When `align_corners = True`, the grid positions depend on the pixel size relative to the input image size, and so the locations sampled by `grid_sample()` will differ for the same input given at different resolutions (that is, after being upsampled or downsampled). The default behavior up to version 1.2.0 was `align_corners = True`. Since then, the default behavior has been changed to `align_corners = False`, in order to bring it in line with the default for `interpolate()`.

⚠ WARNING

Warning When `align_corners = True`, 2D affine transforms on 1D data and 3D affine transforms on 2D data (that is, when one of the spatial dimensions has unit size) are ill-defined, and not an intended use case. This is not a problem when `align_corners = False`. Up to version 1.2.0, all grid points along a unit dimension were considered arbitrarily to be at -1. From version 1.3.0, under `align_corners = True` all grid points along a unit dimension are considered to be at 0 (the center of the input image).

Detailed Documentation

Documentation

Generate 2D or 3D flow field (sampling grid), given a batch of affine matrices theta.

This function is often used in conjunction with `grid_sample()` to build Spatial Transformer Networks .

theta (Tensor) – input batch of affine matrices with shape $(N \times 2 \times 3N \times 2 \times 3N \times 2 \times 3)$ for 2D or $(N \times 3 \times 4N \times 3 \times 4N \times 3 \times 4)$ for 3D

size (torch.Size) – the target output image size. $(N \times C \times H \times W \times N \times C \times H \times W \times N \times C \times H \times W)$ for 2D or $(N \times C \times D \times H \times W \times N \times C \times D \times H \times W \times N \times C \times D \times H \times W)$ for 3D) Example: `torch.Size((32, 3, 24, 24))`

align_corners (bool, optional) – if True, consider -1 and 1 to refer to the centers of the corner pixels rather than the image corners. Refer to `grid_sample()` for a more complete description. A grid generated by `affine_grid()` should be passed to `grid_sample()` with the same setting for this option. Default: False

output Tensor of size $(N \times H \times W \times 2N \times H \times W \times 2N \times H \times W \times 2)$

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